Height Growth Equations for CONUS: Douglas-fir

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Data

We took the FIA height growth data subset for Douglas-fir, filtered out stray (likely keypunch errors) states (NY, PA, IL, and MI), FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 141640 growth observations.

The observations are distributed among L3 Codes as follows:

NA_L3NAME	NA_L3CODE	n
Cascades	6.2.7	33303
Coast Range	7.1.8	23832
Klamath Mountains	6.2.11	16197
Columbia Mountains/Northern Rockies	6.2.3	15792
Middle Rockies	6.2.10	10533
Blue Mountains	6.2.9	7589
North Cascades	6.2.5	6849
Southern Rockies	6.2.14	5023
Idaho Batholith	6.2.15	4946
Eastern Cascades Slopes and Foothills	6.2.8	3582
Sierra Nevada	6.2.12	2674
Strait of Georgia/Puget Lowland	7.1.7	2620
Canadian Rockies	6.2.4	1968
Wasatch and Uinta Mountains	6.2.13	1328
Arizona/New Mexico Mountains	13.1.1	1271
Willamette Valley	7.1.9	1228
Colorado Plateaus	10.1.6	766
Northwestern Great Plains	9.3.3	679
Northern Basin and Range	10.1.3	491

NA_L3NAME	NA_L3CODE	n
California Coastal Sage, Chaparral, and Oak Woodlands	11.1.1	464
Columbia Plateau	10.1.2	284
Central Basin and Range	10.1.5	73
Madrean Archipelago	12.1.1	48
Northwestern Glaciated Plains	9.3.1	38
Wyoming Basin	10.1.4	35
Southwestern Tablelands	9.4.3	16
Arizona/New Mexico Plateau	10.1.7	10
Snake River Plain	10.1.8	1

Height Growth Prediction

We fit the equation:

$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl + \beta_5 PPT_{at} + \beta_6 PPT_{sm} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

to the all 141640 growth observations, estimating β_0 through β_6. Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- PPT_{at} = autumn precipitation (mm)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

ht_{max} was estimated by finding the maximum height observed in the data set for Douglas-fir. In our data that value is: 303 feet.

The parameter estimates and fit statistics are:

	Coef.	Std. error	t-stat.	p
b1	0.0135827	0.0001122	121.05392	0
b2	1.8434663	0.0156232	117.99566	0
b3	0.2288503	0.0061124	37.44040	0
b4	0.0017463	0.0000323	54.06278	0
b5	0.0008566	0.0000121	71.00819	0
b6	-0.0007450	0.0000656	-11.36042	0
b7	0.0266645	0.0002870	92.89148	0

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Residual Standard Error: 6.8909949177029 on 141633 degrees of freedom. AIC =  948757.3
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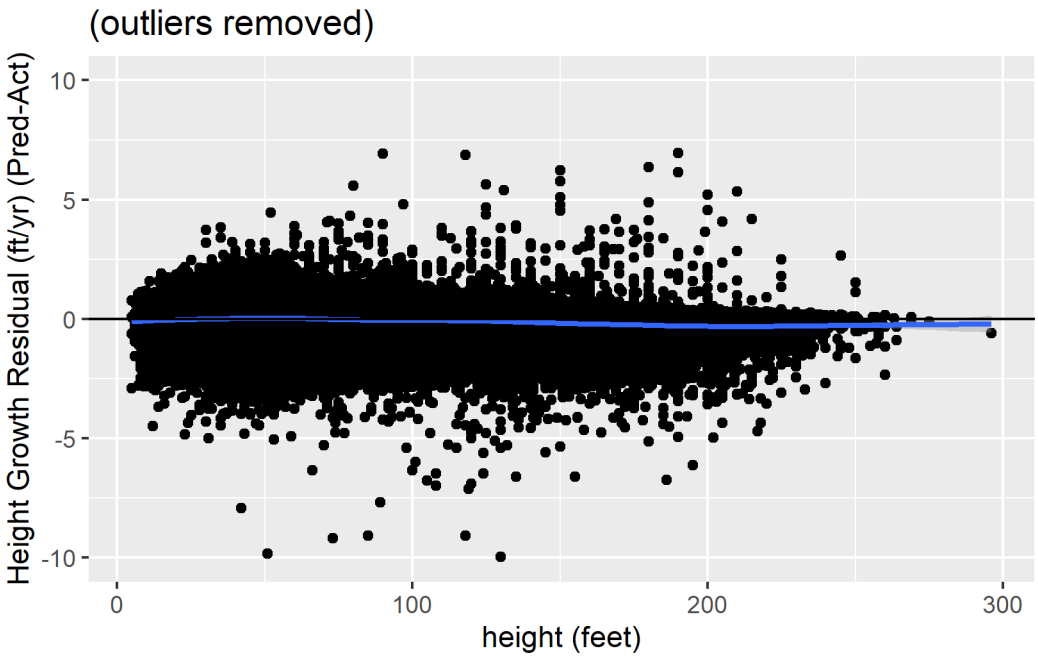
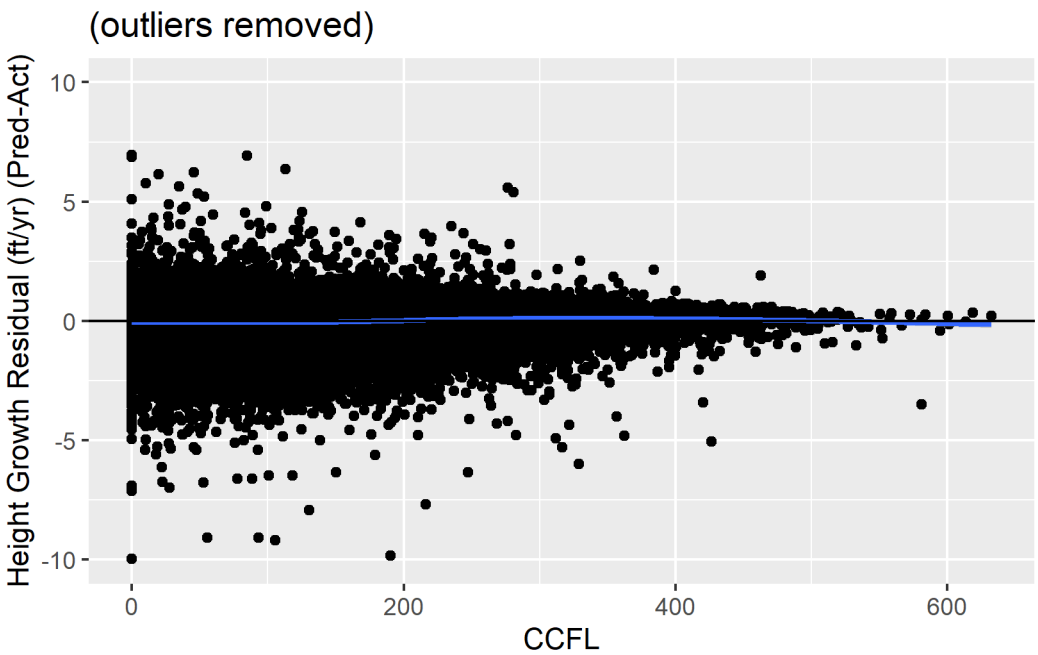
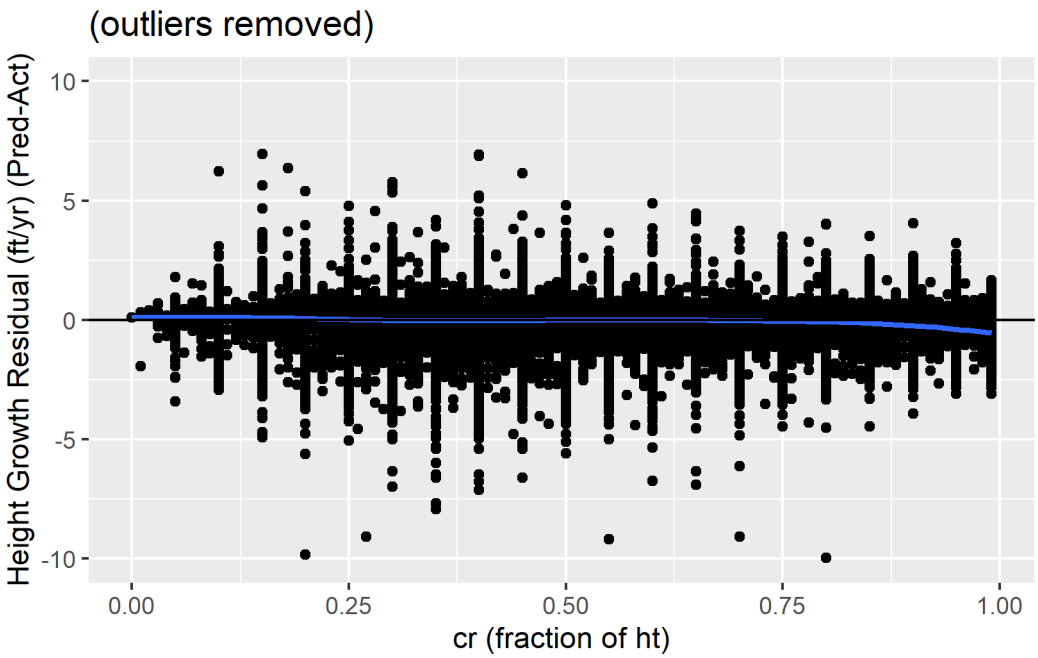
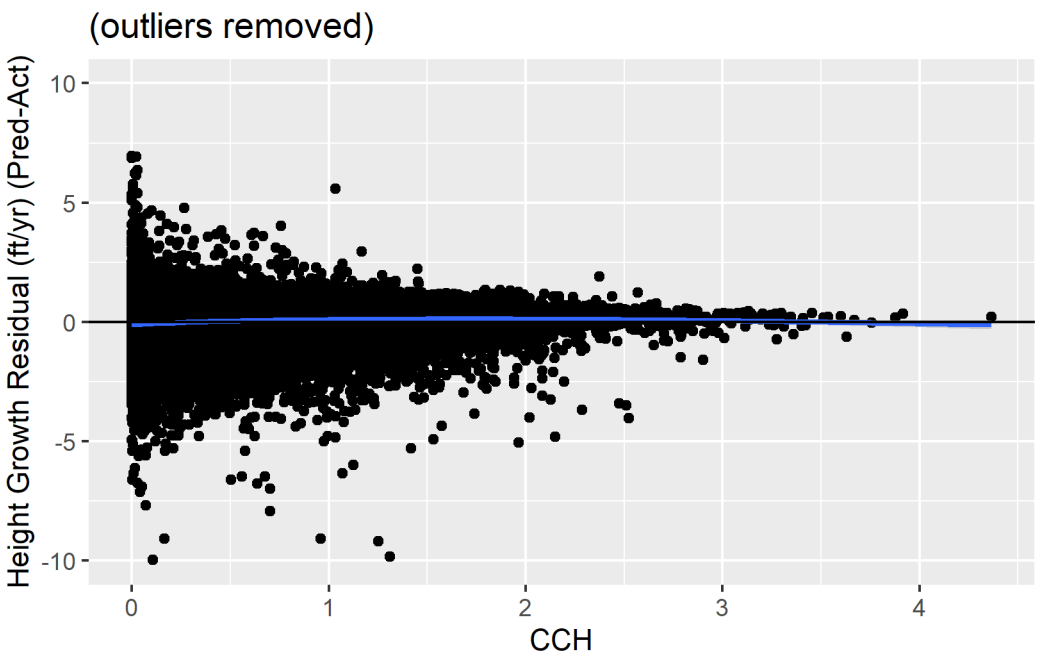
Residual Analysis for Equation 1

Mensurational Variables






(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0.00
0.25
0.50
0.75
1.00
cr (fraction of ht)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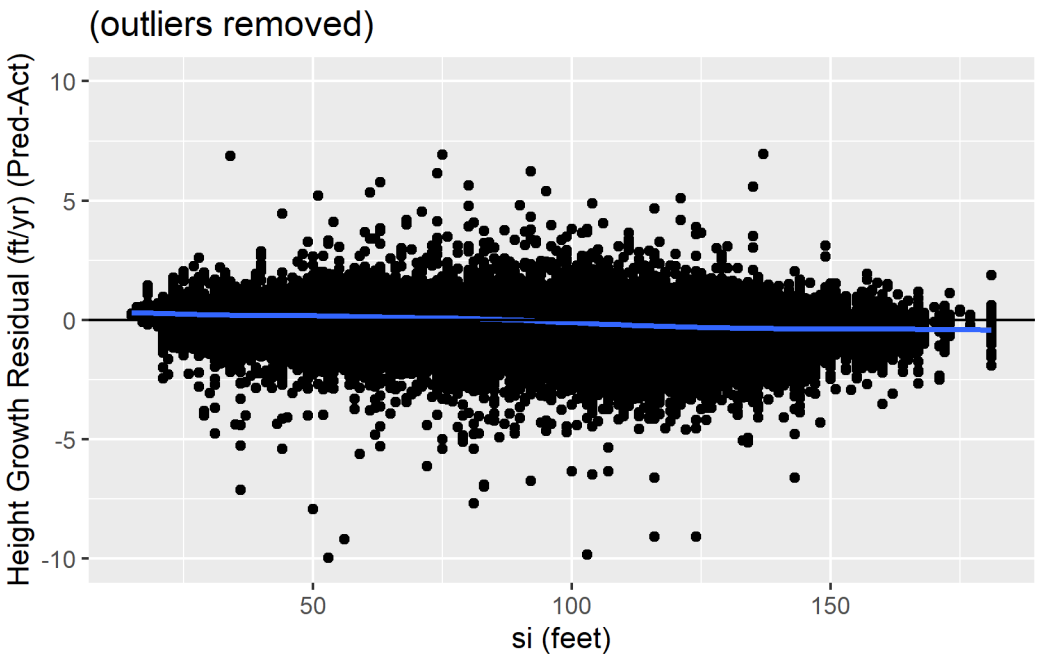
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
1
2
3
4
CCH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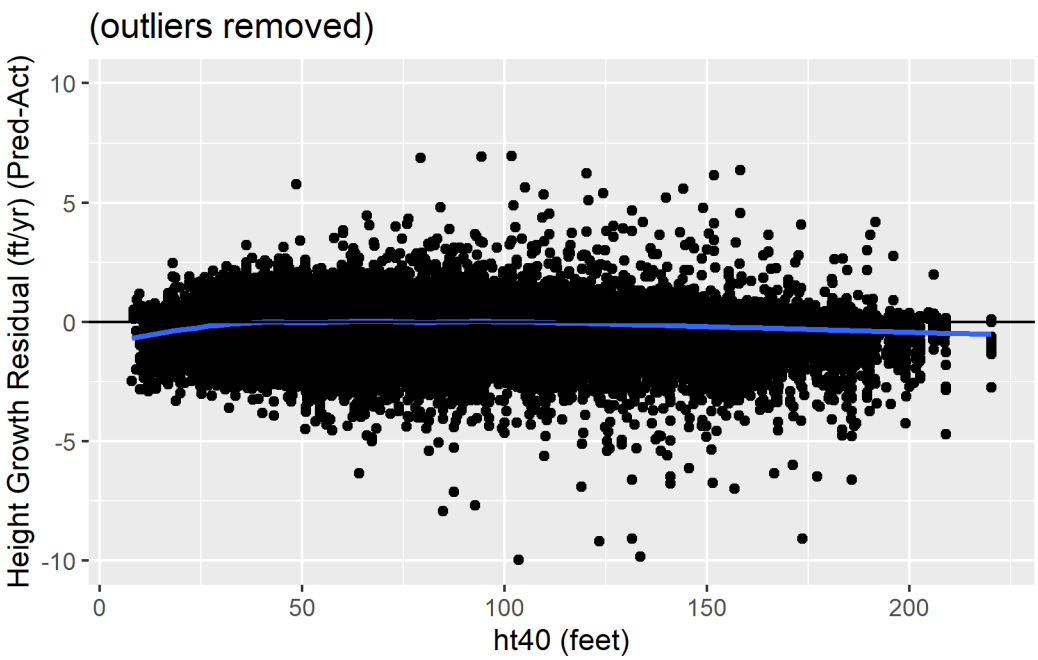
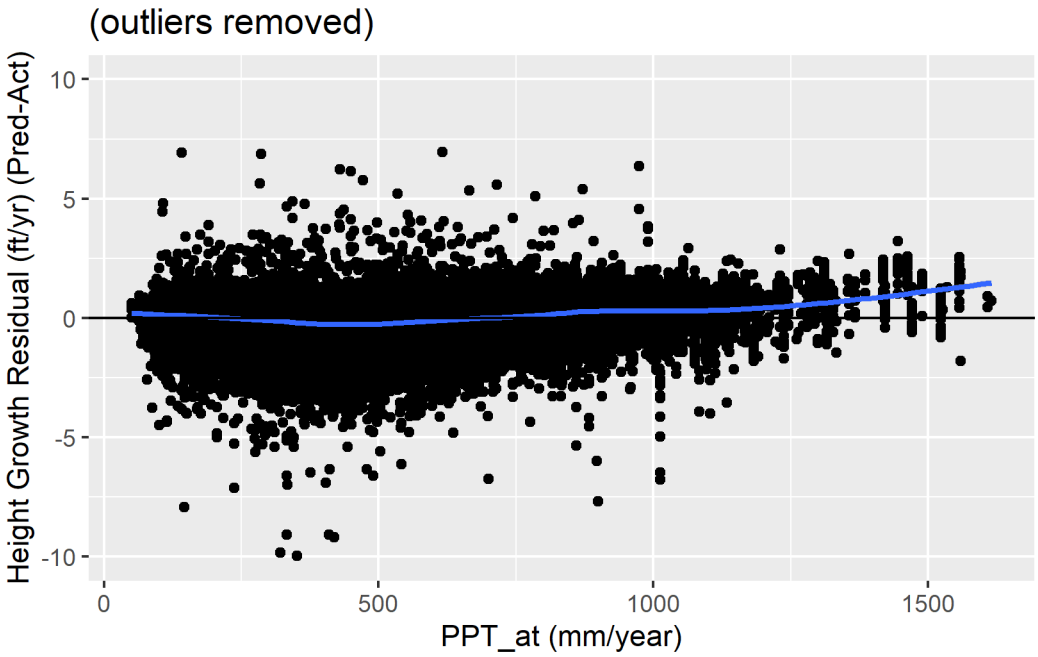
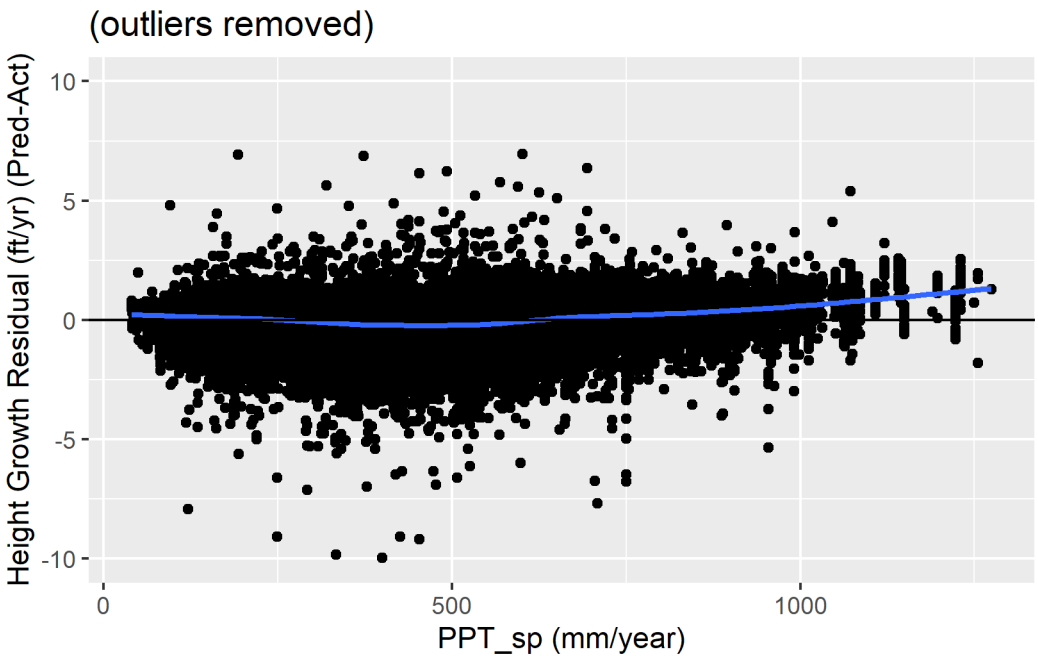
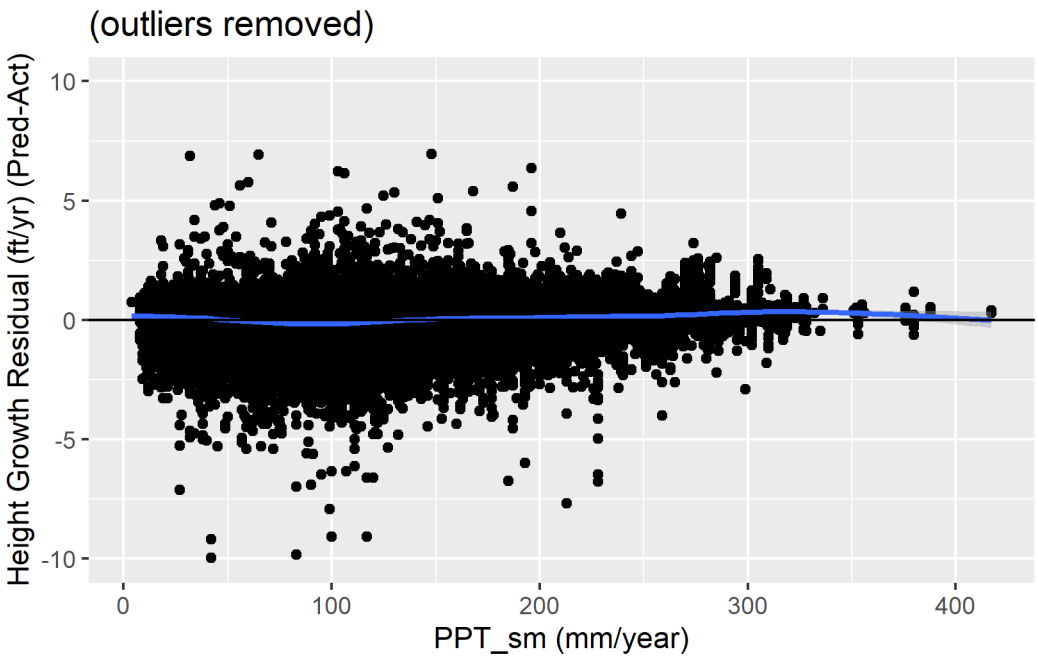
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
50
100
150
si (feet)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
50
100
150
200
ht40 (feet)

Environmental and Climate Variables



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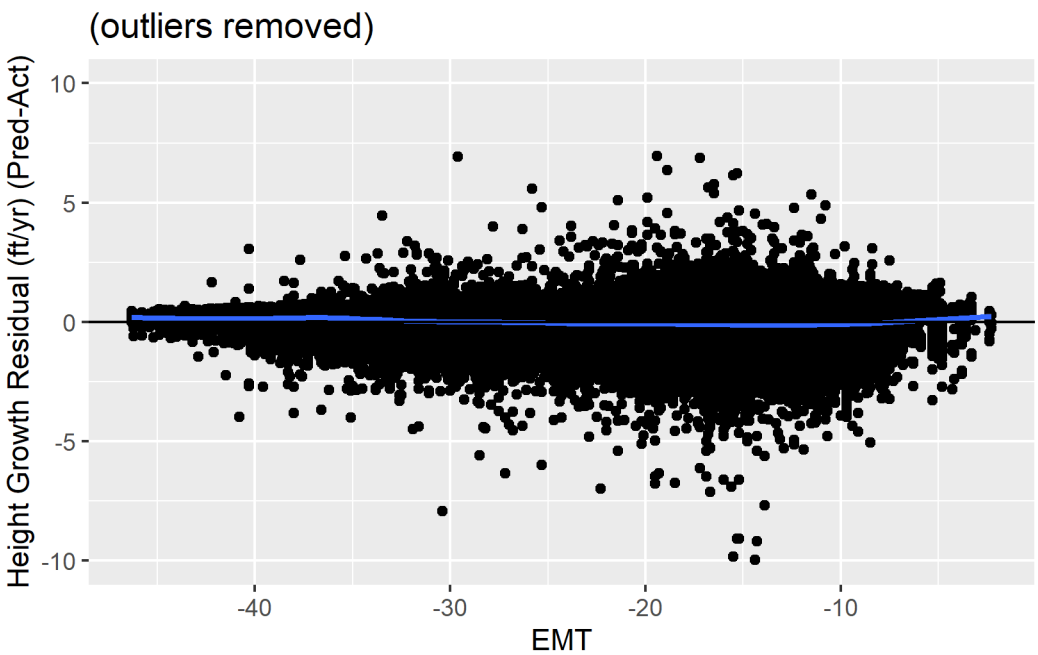
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
100
200
300
400
PPT_sm (mm/year)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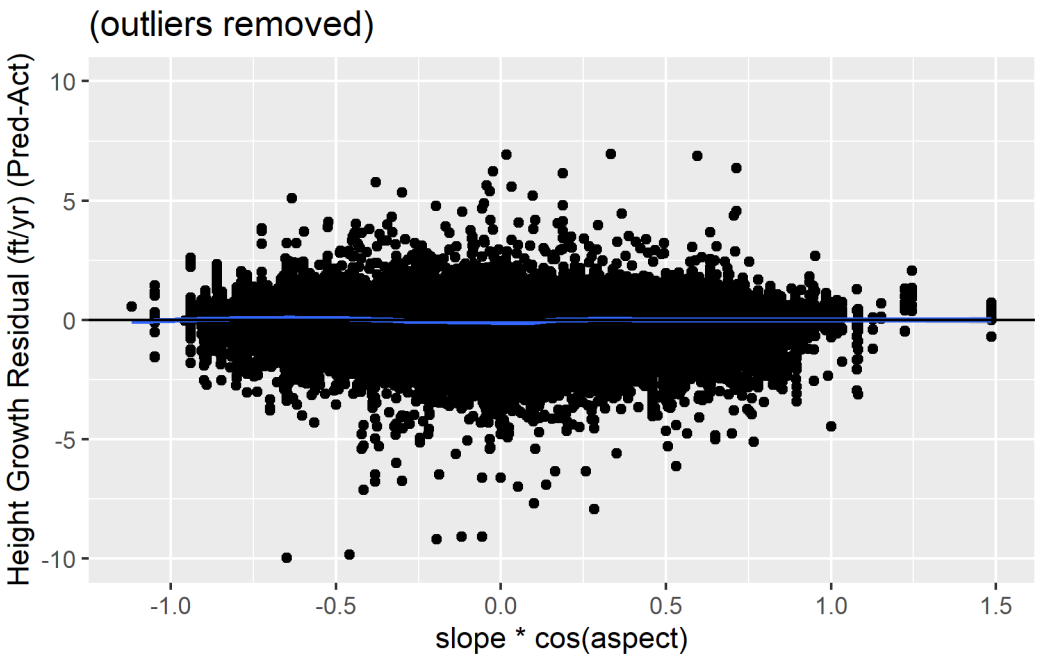
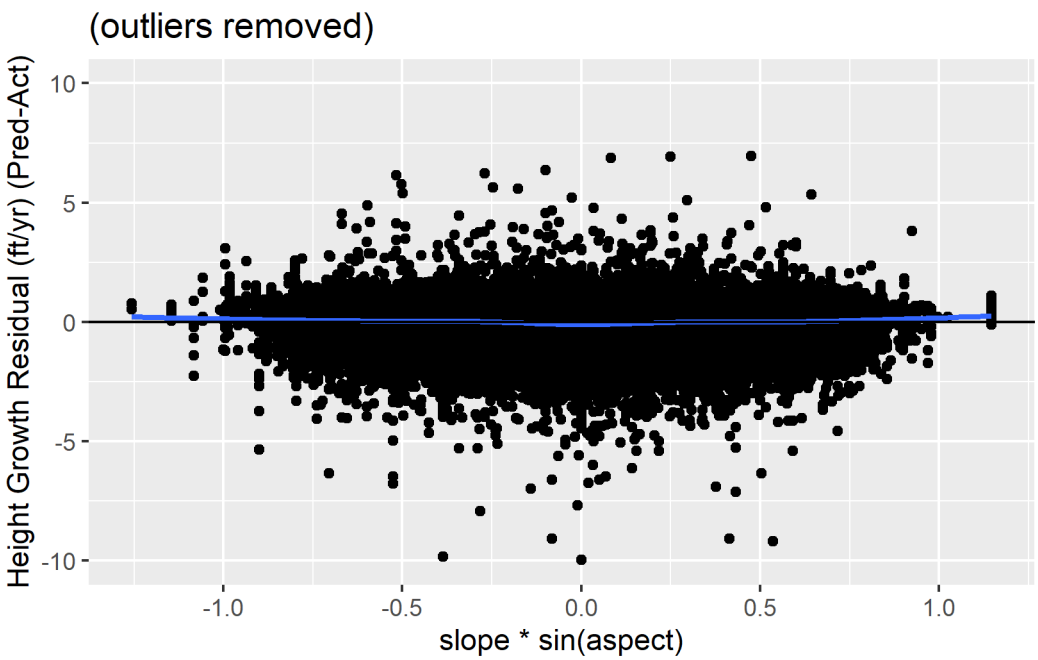
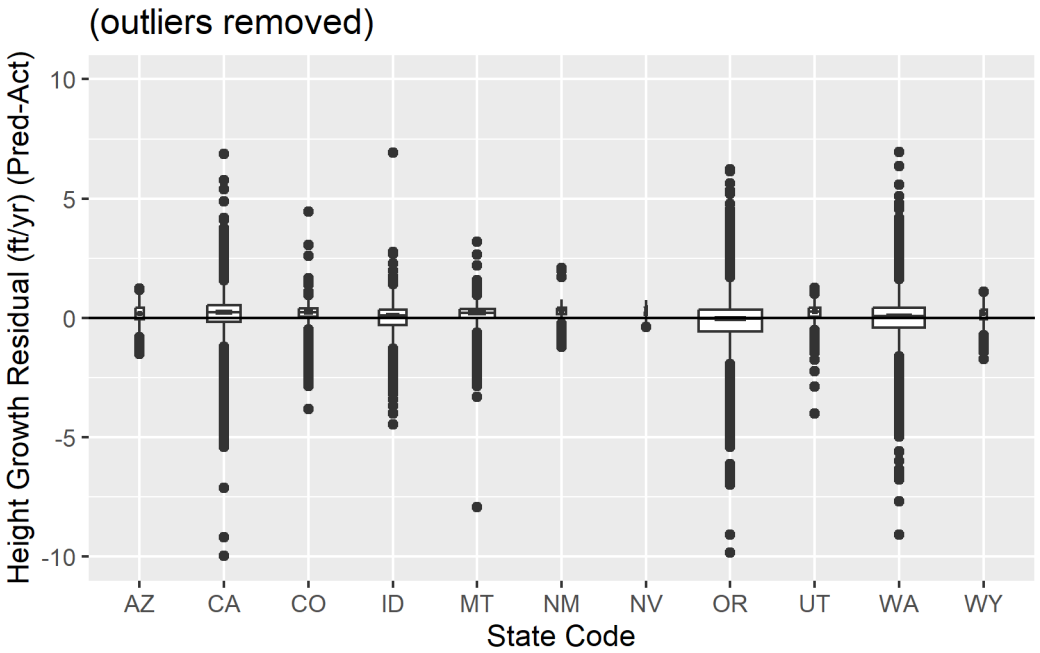
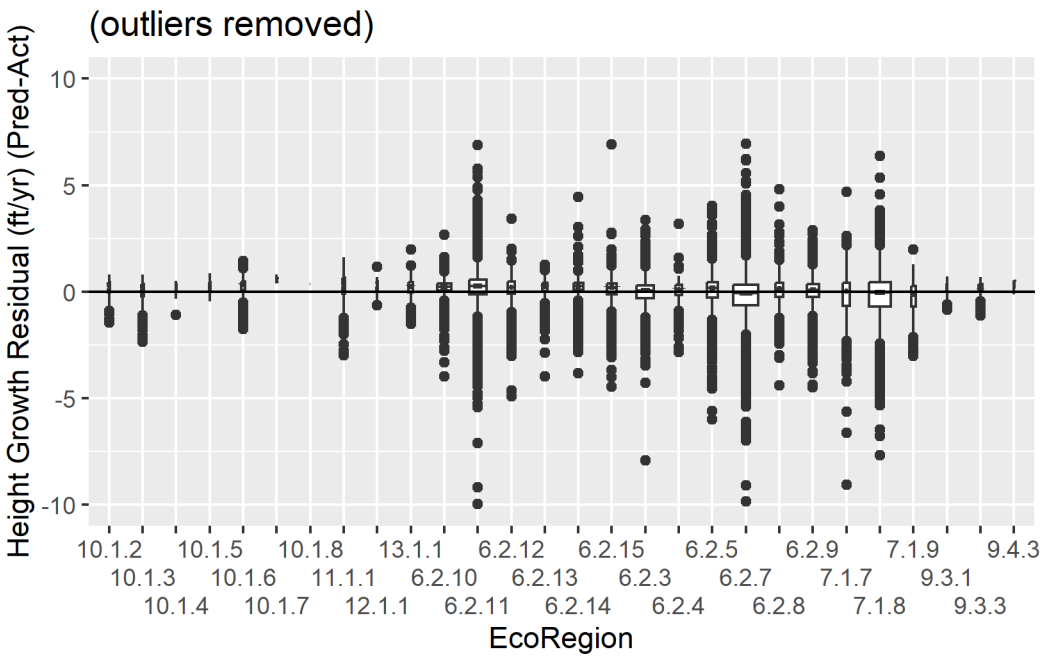
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-40
-30
-20
-10
EMT

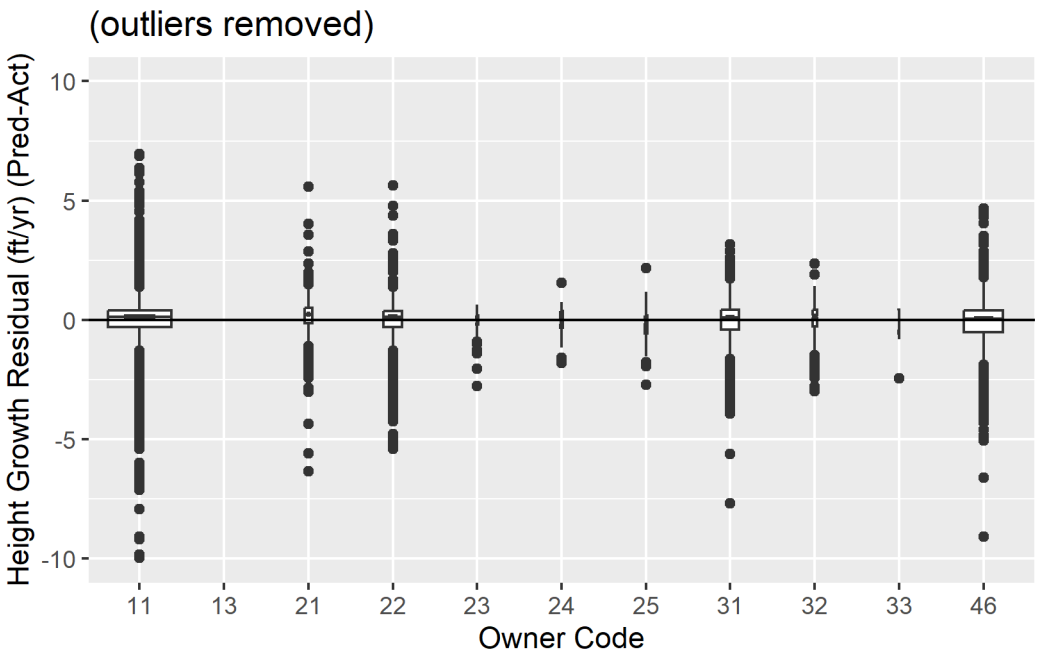

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1.0
-0.5
0.0
0.5
1.0
1.5
slope * cos(aspect)

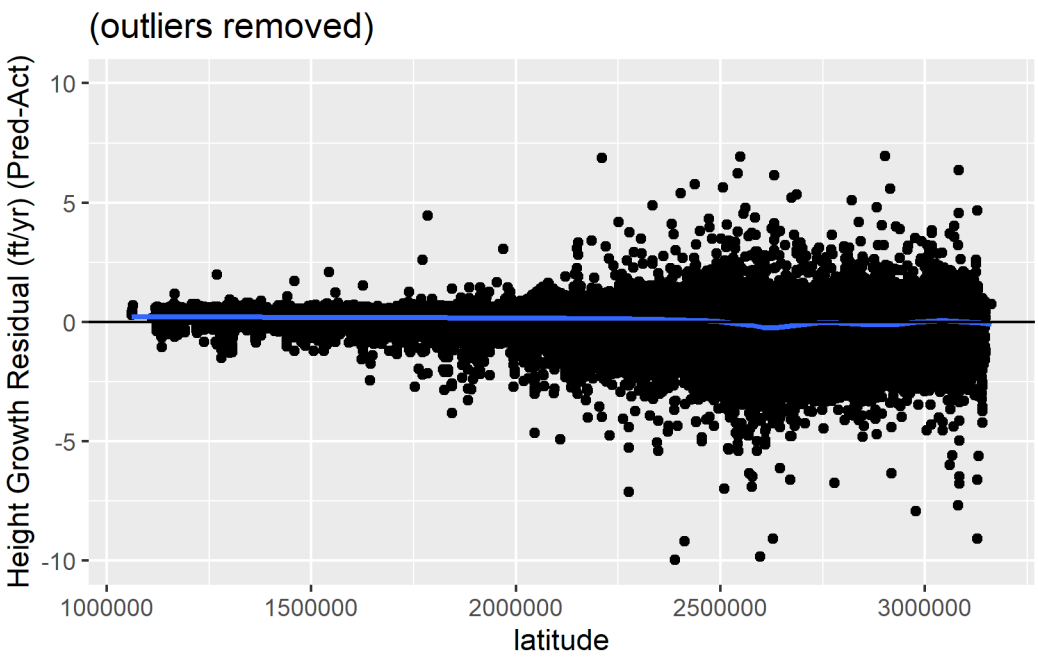

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1.0
-0.5
0.0
0.5
1.0
slope * sin(aspect)

Location Variables






(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
11
13
21
22
23
24
25
31
32
33
46
Owner Code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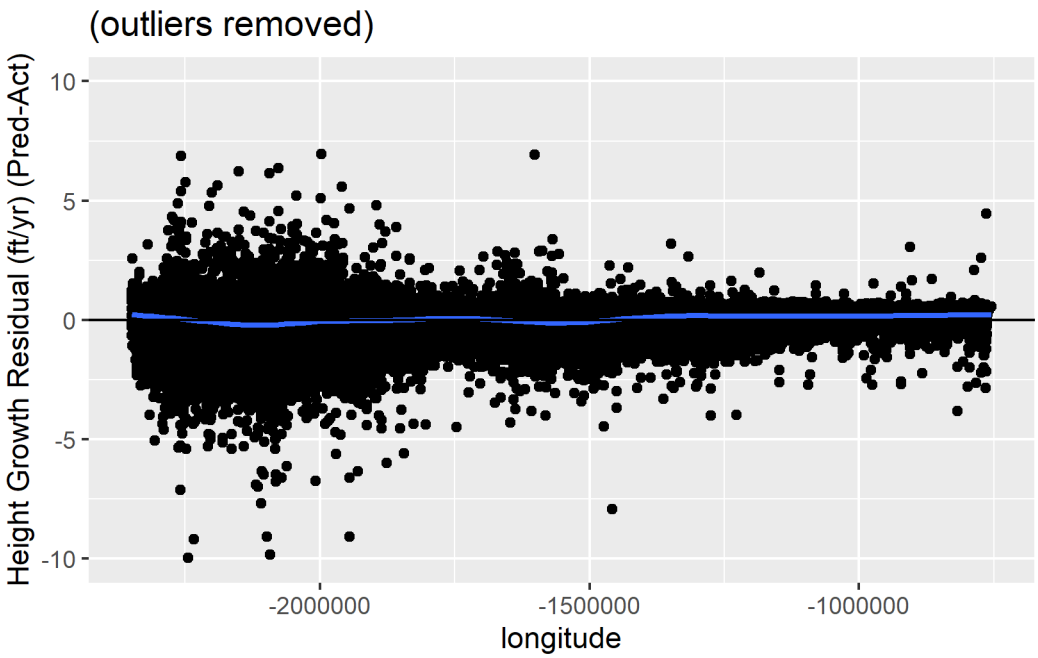
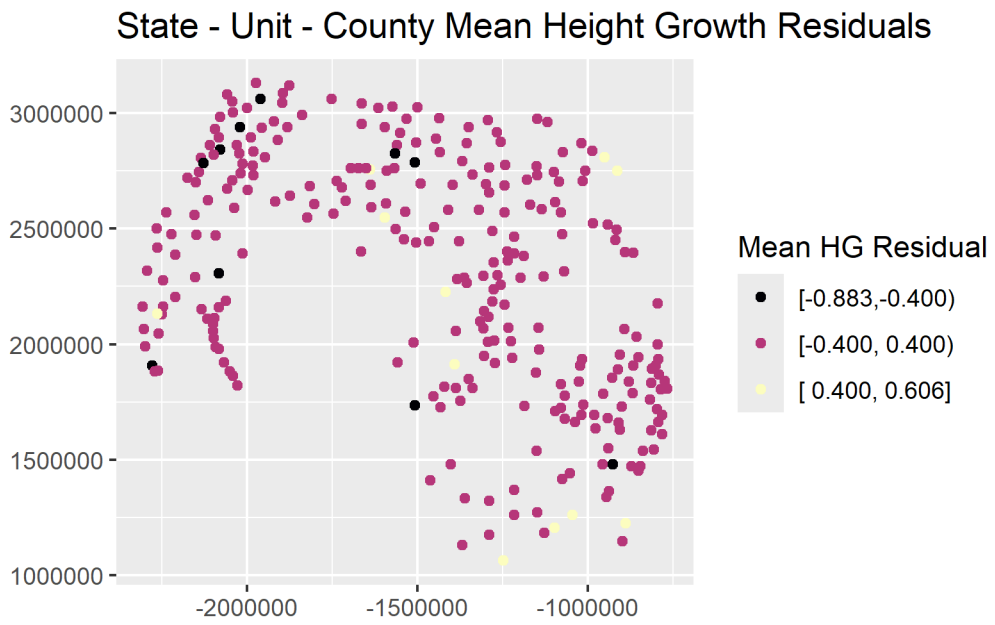
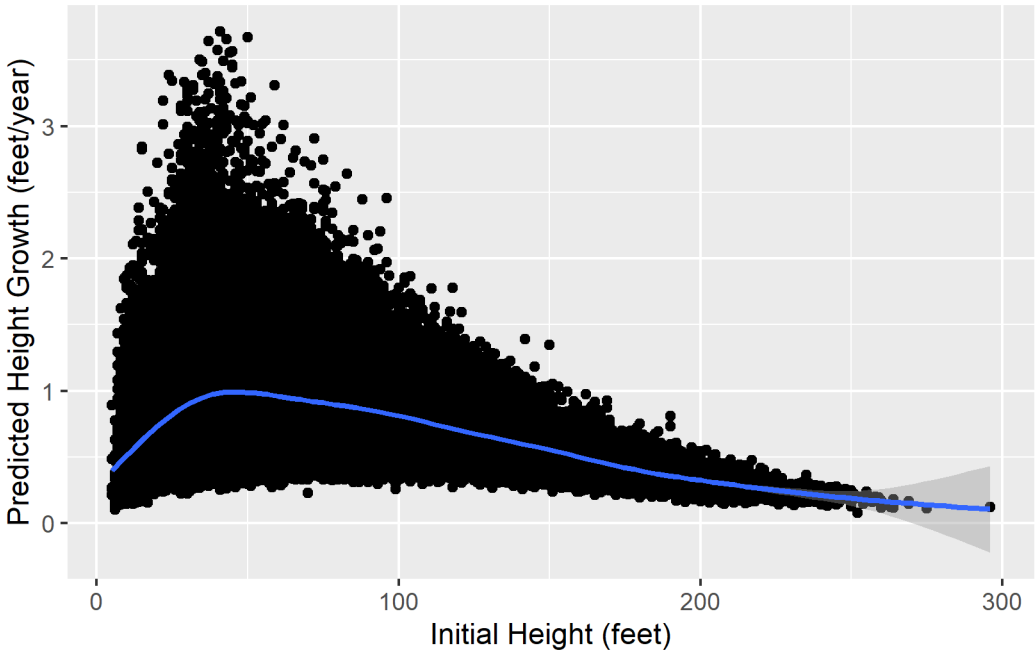
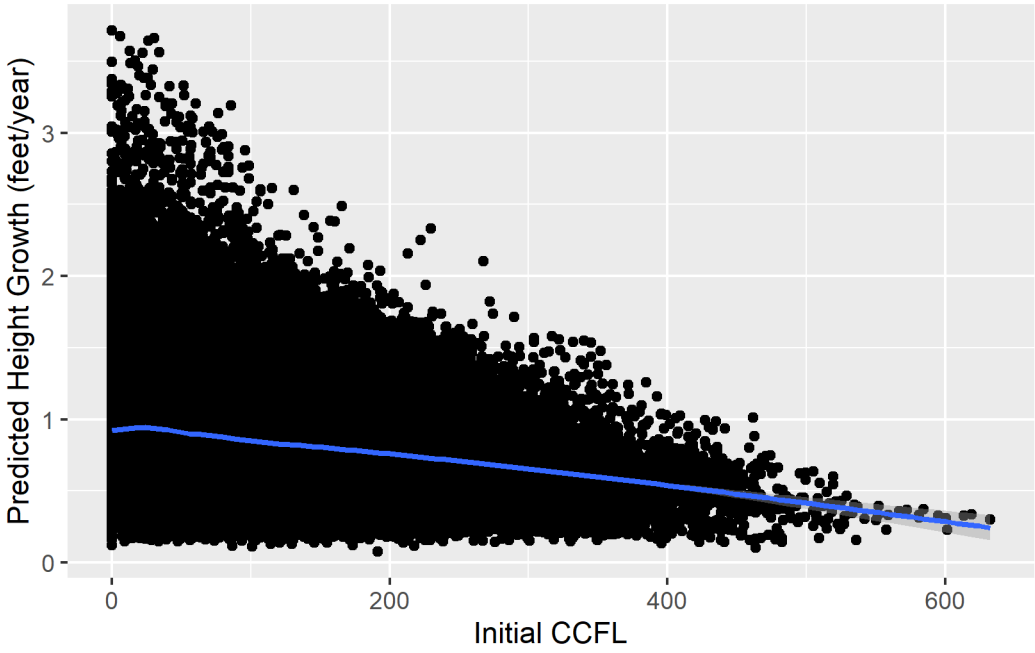
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
1000000
1500000
2000000
2500000
3000000
latitude

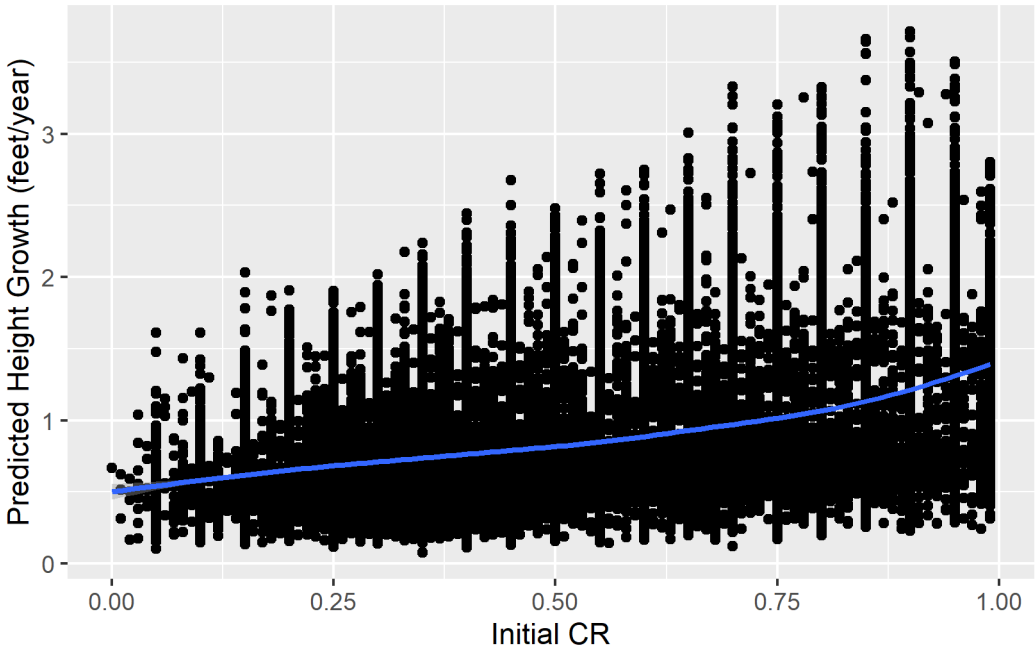

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-2000000
-1500000
-1000000
longitude

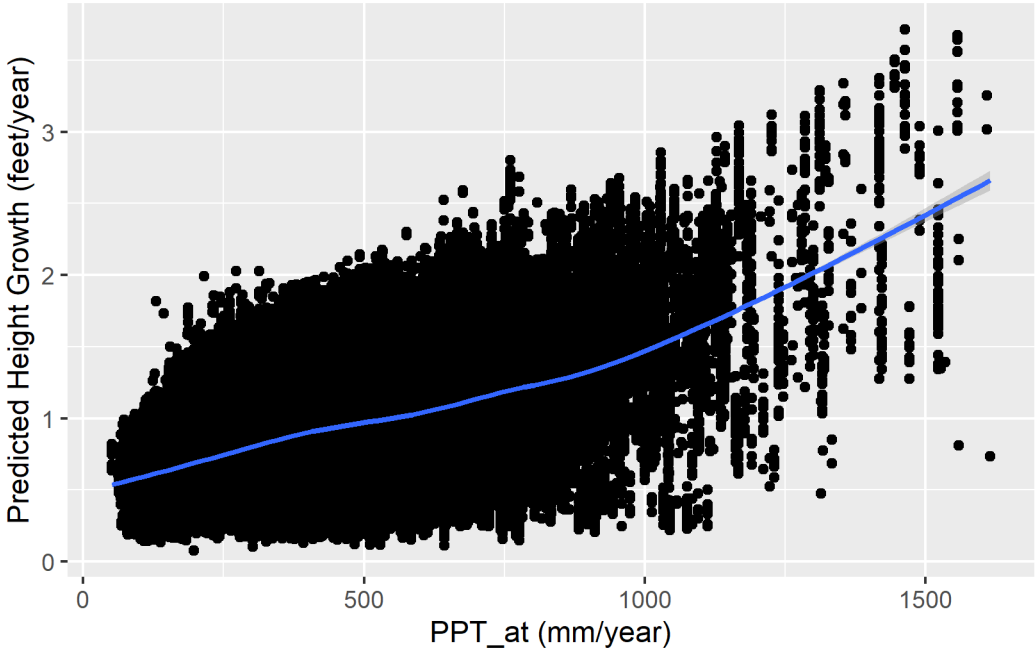

State - Unit - County Mean Height Growth Residuals
3000000
2500000
2000000
1500000
1000000
-2000000
-1500000
-1000000
Mean HG Residual
[-0.883,-0.400)
[-0.400, 0.400)
[0.400, 0.606]

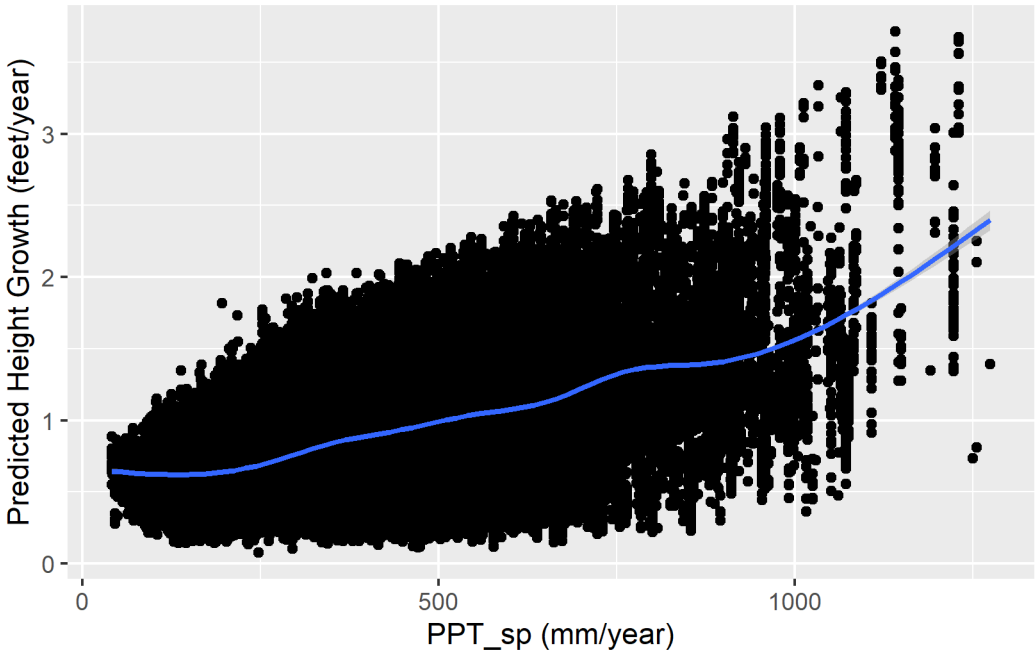
Prediction Performance

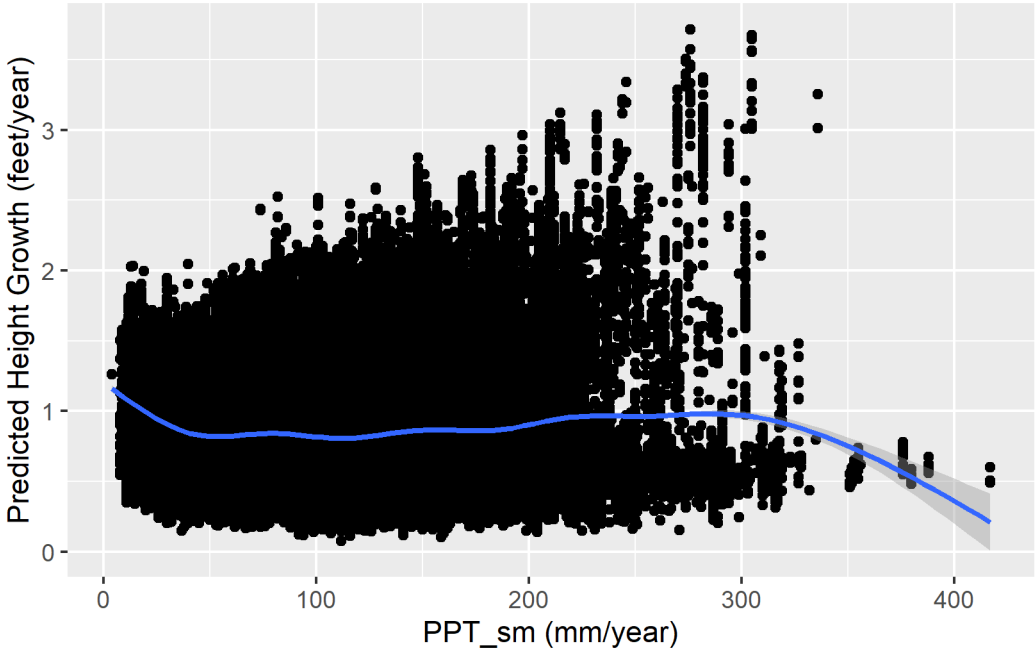


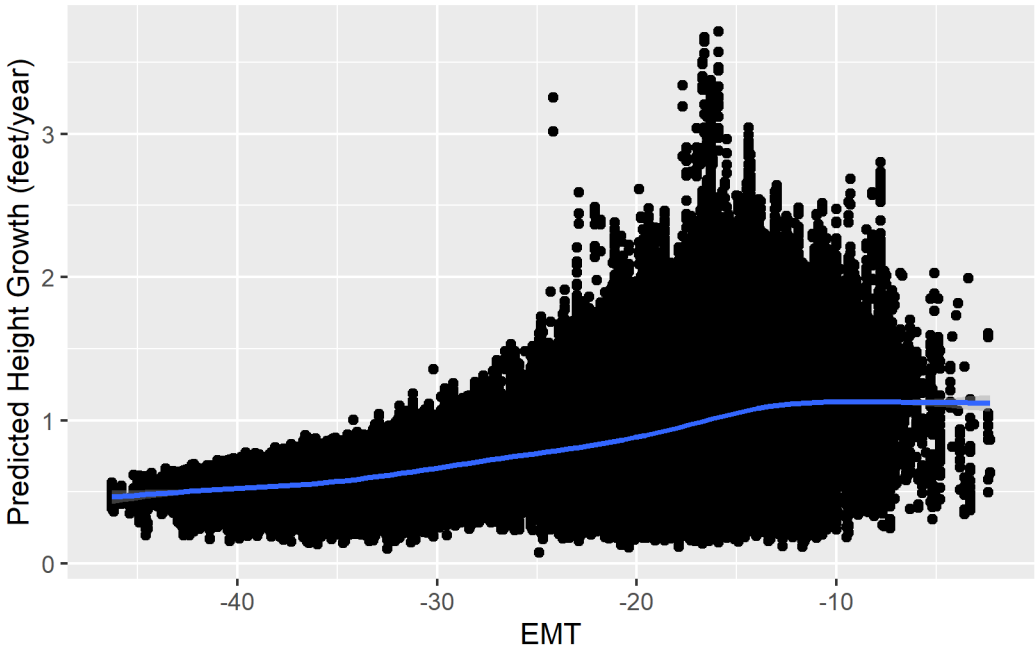



Predicted Height Growth (feet/year)
0
1
2
3
0.00
0.25
0.50
0.75
1.00
Initial CR


Predicted Height Growth (feet/year)
0
1
2
3
0
500
1000
1500
PPT_at (mm/year)


Predicted Height Growth (feet/year)
0
1
2
3
0
500
1000
PPT_sp (mm/year)


Predicted Height Growth (feet/year)
0
1
2
3
0
100
200
300
400
PPT_sm (mm/year)


Predicted Height Growth (feet/year)
0
1
2
3
-40
-30
-20
-10
EMT